

TONOPAH, NV, USA

WMO: 724855

Lat: 38.051N

Lon: 117.090W

Elev: 5395

StdP: 12.05

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23153

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.2	15.4	-8.4	4.3	32.4	-4.2	5.4	35.3	31.7	36.5	27.9	35.2	8.9	0	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.8	95.6	59.2	93.4	58.4	91.1	57.4	63.2	85.6	61.5	83.6	60.2	83.0	11.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.1	85.0	67.8	54.1	76.3	68.9	50.8	67.2	70.5	31.2	85.0	29.9	82.7	28.9	82.9	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.7	21.0	3.2	99.7	4.8	1.7	-0.3	101.0	-3.1	101.9	-5.8	102.9	-9.3	104.1
			1.8	65.6	4.2	2.6	-1.3	67.5	-3.8	69.0	-6.2	70.5	-9.2	72.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	34.2	36.9	43.9	49.6	58.7	69.3	75.7	73.6	65.4	53.0	41.4	32.5	(6)
	DBStd	16.36	7.28	7.28	7.11	7.70	7.82	7.15	4.32	4.04	6.09	7.08	7.53	6.89	(7)
	HDD50	2052	489	367	210	101	19	1	0	0	2	51	270	542	(8)
	HDD65	5278	954	787	654	463	221	42	0	2	66	373	709	1007	(9)
	CDD50	3126	0	1	21	88	289	580	797	730	464	145	11	0	(10)
	CDD65	874	0	0	0	0	25	170	332	267	78	2	0	0	(11)
	CDH74	13976	0	0	3	60	654	2867	4897	3927	1442	124	2	0	(12)
	CDH80	6632	0	0	0	4	176	1350	2649	1962	477	14	0	0	(13)
(14) Wind	WSAvg	9.7	8.8	9.8	10.6	11.8	10.9	10.3	9.1	8.9	9.0	9.3	8.9	8.8	(14)
(15) Precipitation	PrecAvg	5.3	0.4	0.6	0.5	0.4	0.6	0.4	0.4	0.5	0.6	0.4	0.4	0.2	(15)
	PrecMax	10.6	1.1	2.7	2.4	1.7	2.0	1.7	1.7	2.2	2.0	2.2	1.7	1.0	(16)
	PrecMin	2.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	1.9	0.3	0.7	0.6	0.4	0.6	0.5	0.5	0.5	0.6	0.5	0.4	0.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.3	65.2	73.4	79.9	89.6	97.0	99.1	96.6	90.5	81.8	70.5	57.3	(19)
		MCWB	41.8	43.8	47.2	50.7	56.7	58.5	60.5	60.5	56.4	53.1	47.3	41.5	(20)
	2%	DB	55.3	60.3	68.4	75.3	84.5	93.5	96.7	94.2	87.7	77.4	66.0	53.3	(21)
		MCWB	39.8	41.9	45.0	48.3	53.8	57.3	59.9	59.1	55.7	51.1	45.2	39.1	(22)
	5%	DB	51.7	55.9	64.5	71.7	81.1	90.6	94.4	91.9	85.5	73.6	61.1	49.7	(23)
		MCWB	38.7	40.0	43.4	46.7	52.1	56.5	58.8	58.0	54.8	48.8	43.0	37.3	(24)
	10%	DB	47.6	51.1	60.6	67.7	77.3	88.1	92.1	90.0	82.6	70.0	56.3	45.6	(25)
		MCWB	36.7	37.9	41.7	45.1	50.3	55.3	58.0	57.0	54.2	47.3	40.6	35.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	45.6	48.3	51.6	58.8	63.8	66.8	65.4	61.5	55.4	49.4	44.3	(27)
		MCDB	53.6	59.9	68.1	76.6	86.9	90.8	89.2	90.2	80.0	72.0	65.4	52.9	(28)
	2%	WB	42.2	43.3	46.1	49.5	55.2	60.0	63.6	62.4	59.7	52.8	46.3	41.4	(29)
		MCDB	50.1	57.6	65.1	72.9	80.1	89.6	85.8	85.1	79.1	72.0	62.5	48.8	(30)
	5%	WB	40.1	41.2	44.3	47.6	53.1	57.9	62.1	60.8	57.9	50.7	44.0	38.8	(31)
		MCDB	48.5	52.7	62.4	69.1	77.7	87.2	83.8	83.3	78.0	70.1	59.3	47.3	(32)
	10%	WB	37.8	39.0	42.4	45.8	51.2	56.0	60.8	59.5	55.9	48.6	41.6	36.2	(33)
		MCDB	46.7	49.0	58.8	66.0	74.5	84.3	82.8	82.3	76.8	66.8	54.4	44.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	23.0	23.2	27.0	27.3	28.5	31.1	31.8	31.9	30.5	28.7	26.2	23.4	(35)
		MCDBR	29.3	30.7	32.7	33.2	33.6	34.1	33.8	33.3	32.9	33.5	32.3	28.8	(36)
	5% WB	MCWBR	17.1	16.8	16.4	15.9	15.4	15.4	14.4	14.4	15.0	17.0	17.6	17.8	(37)
		MCWBR	24.2	27.0	30.8	31.3	32.1	33.3	29.3	30.2	29.6	30.4	29.9	25.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.238	0.245	0.254	0.287	0.300	0.298	0.316	0.309	0.283	0.259	0.242	0.236	(40)
		taud	2.651	2.636	2.611	2.479	2.474	2.498	2.493	2.492	2.565	2.640	2.658	2.662	(41)
	Ebn at Noon		311	320	324	315	310	310	303	304	309	310	307	305	(42)
		Edh at Noon	21	24	28	34	35	34	34	33	29	24	21	19	(43)
(44) All-Sky Solar Radiation	RadAvg		867	1170	1645	2082	2354	2672	2448	2275	1936	1418	1007	759	(44)
	RadStd		84	116	95	109	180	163	131	128	77	93	43	65	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (7 neighbors)	+0.90	N/A	-1.84	N/A	N/A	N/A	N/A	N/A	+234	+115	(47)

Nomenclature: See separate page